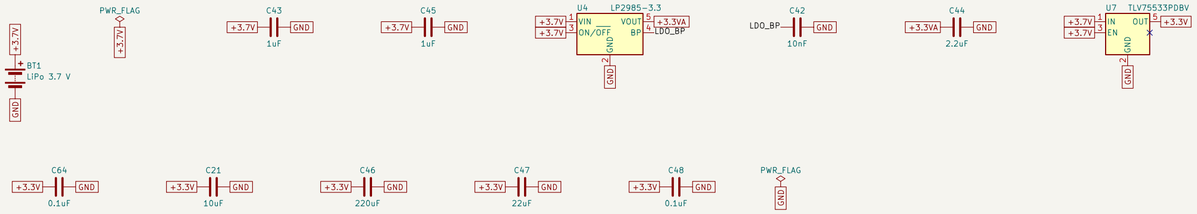
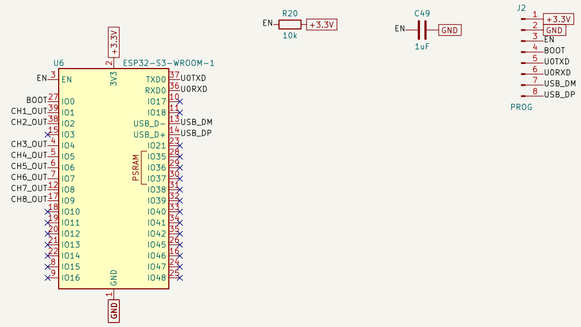
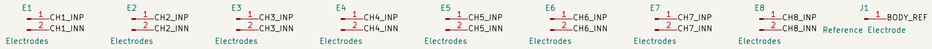
Power: battery, analog LDO U4 -> +3.3VA (front ends), digital LDO U7 -> +3.3V (ESP32-S3)



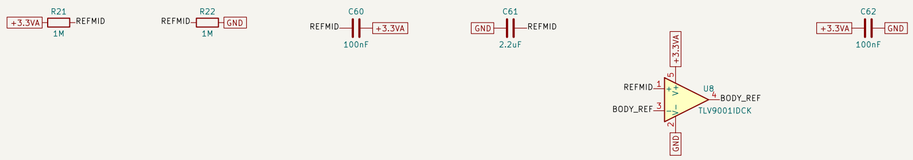
Microcontroller: ESP32-S3-WROOM-1 (ADC1 on IO1, 2, 4 to 9), EN reset RC, programming header



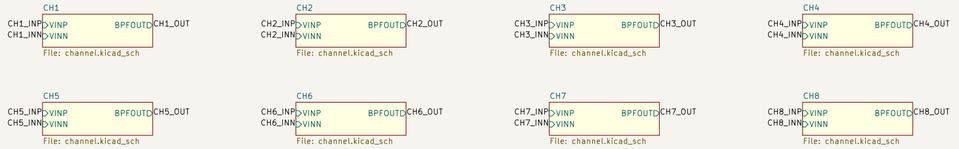
Electrodes: channel pairs E1 to E8 (pin 1 +, pin 2 -), reference electrode J1



Body reference: J1 held at VCC/2 = 1.65 V (Patchy's reference block, buffered by U8)



Front-end channels: sheet channel.kicad_sch, eight instances



Sections from the client's base design (open-semg-8), with the supply fixes of the 28 Oct audit: U4 feeds only +3.3VA, U7 only +3.3V, both always on; one ground.
Added: J2 programming header; the body-reference buffer (U8). Each channel's BPFOUT goes to an ESP32-S3 ADC1 pin (ATTEN3, 0 to 2.9 V): CH1 IO1, CH2 IO2, CH3 IO4, CH4 IO5, CH5 IO6, CH6 IO7, CH7 IO8, CH8 IO9 (IO3, a strapping pin, is skipped).
Boot: hold J2 pin 4 (IO0) low at reset for download mode. The front ends were simulated by the design loop; the board-level sections were checked by netlist and ERC only.

Licence CERN-OHL-S-2.0 · DigiKey BOM 2026-11-02
Front end: ambrflow design loop, engine e735a0e, vref-0065
Power, MCU, connectors: client's open-semg-8 (audit fixes 28 Oct)

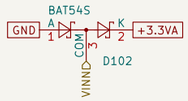
Sheet: /
File: semg8_A2_MCP6004.kicad_sch

Title: sEMG-8 board A2: MCP6004-I/SL front ends

Size: A2	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/9

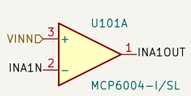
Input: RFI filter and clamps to the rails

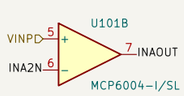


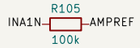


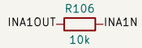


Instrumentation amplifier U1A / U1B: gain 1 + R8/R7 = 11; CMRR needs R5/R6 = R8/R7





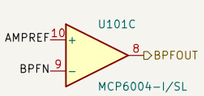


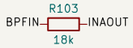






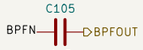
Band-pass into the ADC, U1C: R3/C4 high-pass (4.0 Hz), R4/C5 low-pass (723 Hz), gain R4/R3 = 12.2



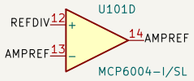


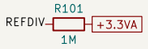






Mid-supply reference: R1/R2 divider (1.055 V), C2/C3 filter, U1D buffer



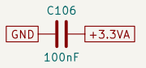




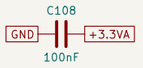


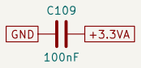


Supply: U1 power pins and bypass











One sEMG channel, placed eight times on the root sheet (CH1 to CH8; references R1xx to R8xx by channel).
Ports: VINP / VINN from the electrode pair, BPFOUT to an ESP32-S3 ADC1 pin. Supply +3.3VA (the LP2985 analog rail), GND.
Simulated: gain 134 V/V at 50 to 100 Hz, 110 V/V at 500 Hz; BPFOUT rests at 1.055 V; CMRR >= 48.6 dB at every corner (test_limits.csv).
U1 is KiCad's generic quad op-amp symbol (LM324 pinout, the same as the fitted part); the fitted part is the Value / MPN field.

Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0065: Muscle BioAmp Patchy v0.2 (CERN-OHL-S-2.0)
ambrflow design loop, engine e735a0e; spec semg8-opamp 73a6d4ada293

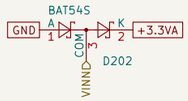
Sheet: /CH1/
File: channel.kicad_sch

Title: sEMG-8 front-end channel (one of eight)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/9

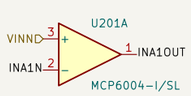
Input: RFI filter and clamps to the rails

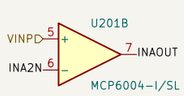




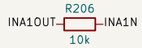


Instrumentation amplifier U1A / U1B: gain 1 + R8/R7 = 11; CMRR needs R5/R6 = R8/R7





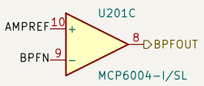




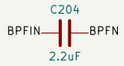




Band-pass into the ADC, U1C: R3/C4 high-pass (4.0 Hz), R4/C5 low-pass (723 Hz), gain R4/R3 = 12.2



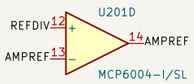




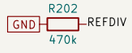


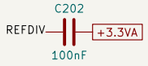


Mid-supply reference: R1/R2 divider (1.055 V), C2/C3 filter, U1D buffer



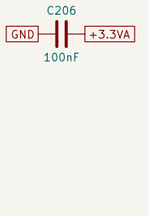






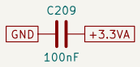


Supply: U1 power pins and bypass









One sEMG channel, placed eight times on the root sheet (CH1 to CH8; references R1xx to R8xx by channel).
Ports: VINP / VINN from the electrode pair, BPFOUT to an ESP32-S3 ADC1 pin. Supply +3.3VA (the LP2985 analog rail), GND.
Simulated: gain 134 V/V at 50 to 100 Hz, 110 V/V at 500 Hz; BPFOUT rests at 1.055 V; CMRR >= 48.6 dB at every corner (test_limits.csv).
U1 is KiCad's generic quad op-amp symbol (LM324 pinout, the same as the fitted part); the fitted part is the Value / MPN field.

Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0065: Muscle BioAmp Patchy v0.2 (CERN-OHL-S-2.0)
ambrflow design loop, engine e735a0e; spec semg8-opamp 73a6d4ada293

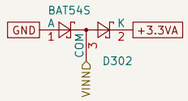
Sheet: /CH2/
File: channel.kicad_sch

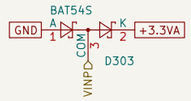
Title: sEMG-8 front-end channel (one of eight)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/9

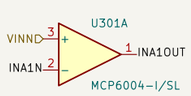
Input: RFI filter and clamps to the rails

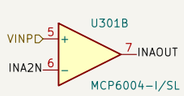






Instrumentation amplifier U1A / U1B: gain 1 + R8/R7 = 11; CMRR needs R5/R6 = R8/R7





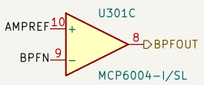








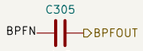
Band-pass into the ADC, U1C: R3/C4 high-pass (4.0 Hz), R4/C5 low-pass (723 Hz), gain R4/R3 = 12.2



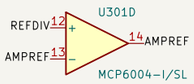




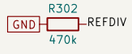


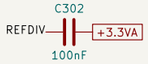


Mid-supply reference: R1/R2 divider (1.055 V), C2/C3 filter, U1D buffer



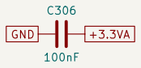








Supply: U1 power pins and bypass











One sEMG channel, placed eight times on the root sheet (CH1 to CH8; references R1xx to R8xx by channel).
Ports: VINP / VINN from the electrode pair, BPFOUT to an ESP32-S3 ADC1 pin. Supply +3.3VA (the LP2985 analog rail), GND.
Simulated: gain 134 V/V at 50 to 100 Hz, 110 V/V at 500 Hz; BPFOUT rests at 1.055 V; CMRR >= 48.6 dB at every corner (test_limits.csv).
U1 is KiCad's generic quad op-amp symbol (LM324 pinout, the same as the fitted part); the fitted part is the Value / MPN field.

Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0065: Muscle BioAmp Patchy v0.2 (CERN-OHL-S-2.0)
ambrflow design loop, engine e735a0e; spec semg8-opamp 73a6d4ada293

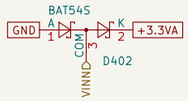
Sheet: /CH3/
File: channel.kicad_sch

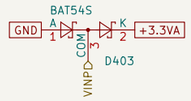
Title: sEMG-8 front-end channel (one of eight)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/9

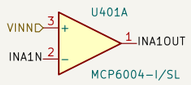
Input: RFI filter and clamps to the rails

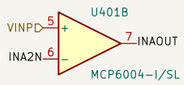






Instrumentation amplifier U1A / U1B: gain 1 + R8/R7 = 11; CMRR needs R5/R6 = R8/R7





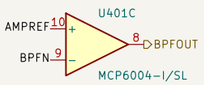








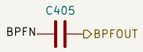
Band-pass into the ADC, U1C: R3/C4 high-pass (4.0 Hz), R4/C5 low-pass (723 Hz), gain R4/R3 = 12.2



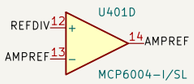








Mid-supply reference: R1/R2 divider (1.055 V), C2/C3 filter, U1D buffer



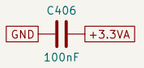








Supply: U1 power pins and bypass











One sEMG channel, placed eight times on the root sheet (CH1 to CH8; references R1xx to R8xx by channel).
Ports: VINP / VINN from the electrode pair, BPFOUT to an ESP32-S3 ADC1 pin. Supply +3.3VA (the LP2985 analog rail), GND.
Simulated: gain 134 V/V at 50 to 100 Hz, 110 V/V at 500 Hz; BPFOUT rests at 1.055 V; CMRR >= 48.6 dB at every corner (test_limits.csv).
U1 is KiCad's generic quad op-amp symbol (LM324 pinout, the same as the fitted part); the fitted part is the Value / MPN field.

Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0065: Muscle BioAmp Patchy v0.2 (CERN-OHL-S-2.0)
ambrflow design loop, engine e735a0e; spec semg8-opamp 73a6d4ada293

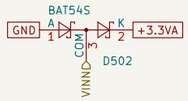
Sheet: /CH4/
File: channel.kicad_sch

Title: sEMG-8 front-end channel (one of eight)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/9

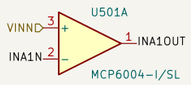
Input: RFI filter and clamps to the rails

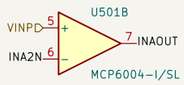






Instrumentation amplifier U1A / U1B: gain 1 + R8/R7 = 11; CMRR needs R5/R6 = R8/R7





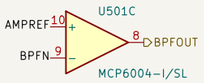








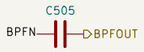
Band-pass into the ADC, U1C: R3/C4 high-pass (4.0 Hz), R4/C5 low-pass (723 Hz), gain R4/R3 = 12.2



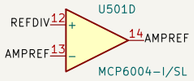








Mid-supply reference: R1/R2 divider (1.055 V), C2/C3 filter, U1D buffer



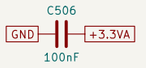


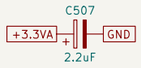


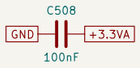




Supply: U1 power pins and bypass











One sEMG channel, placed eight times on the root sheet (CH1 to CH8; references R1xx to R8xx by channel).
Ports: VINP / VINN from the electrode pair, BPFOUT to an ESP32-S3 ADC1 pin. Supply +3.3VA (the LP2985 analog rail), GND.
Simulated: gain 134 V/V at 50 to 100 Hz, 110 V/V at 500 Hz; BPFOUT rests at 1.055 V; CMRR >= 48.6 dB at every corner (test_limits.csv).
U1 is KiCad's generic quad op-amp symbol (LM324 pinout, the same as the fitted part); the fitted part is the Value / MPN field.

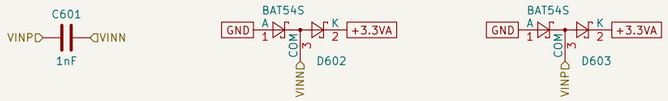
Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0065: Muscle BioAmp Patchy v0.2 (CERN-OHL-S-2.0)
ambrflow design loop, engine e735a0e; spec semg8-opamp 73a6d4ada293

Sheet: /CH5/
File: channel.kicad_sch

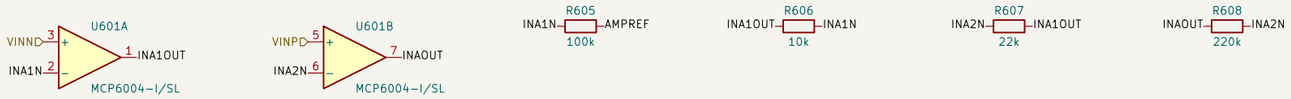
Title: sEMG-8 front-end channel (one of eight)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/9

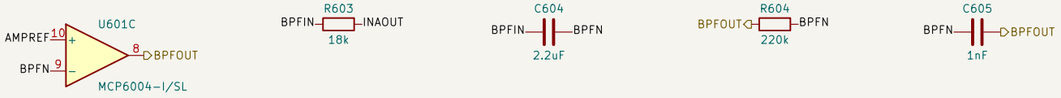
Input: RFI filter and clamps to the rails



Instrumentation amplifier U1A / U1B: gain 1 + R8/R7 = 11; CMRR needs R5/R6 = R8/R7



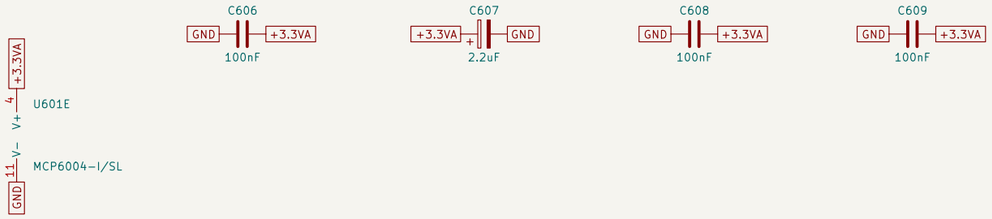
Band-pass into the ADC, U1C: R3/C4 high-pass (4.0 Hz), R4/C5 low-pass (723 Hz), gain R4/R3 = 12.2



Mid-supply reference: R1/R2 divider (1.055 V), C2/C3 filter, U1D buffer



Supply: U1 power pins and bypass



One sEMG channel, placed eight times on the root sheet (CH1 to CH8; references R1xx to R8xx by channel).
Ports: VINP / VINN from the electrode pair, BPFOUT to an ESP32-S3 ADC1 pin. Supply +3.3VA (the LP2985 analog rail), GND.
Simulated: gain 134 V/V at 50 to 100 Hz, 110 V/V at 500 Hz; BPFOUT rests at 1.055 V; CMRR >= 48.6 dB at every corner (test_limits.csv).
U1 is KiCad's generic quad op-amp symbol (LM324 pinout, the same as the fitted part); the fitted part is the Value / MPN field.

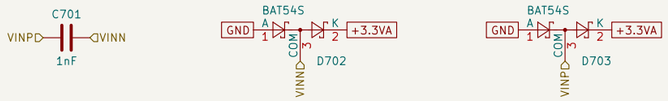
Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0065: Muscle BioAmp Patchy v0.2 (CERN-OHL-S-2.0)
ambrflow design loop, engine e735a0e; spec semg8-opamp 73a6d4ada293

Sheet: /CH6/
File: channel.kicad_sch

Title: sEMG-8 front-end channel (one of eight)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/9

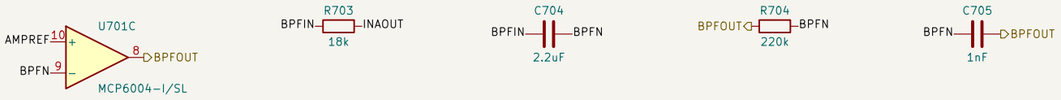
Input: RFI filter and clamps to the rails



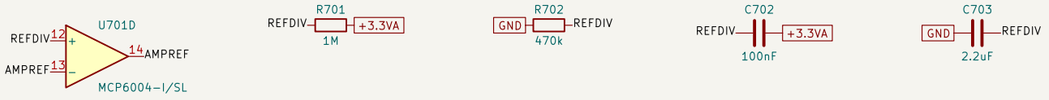
Instrumentation amplifier U1A / U1B: gain 1 + R8/R7 = 11; CMRR needs R5/R6 = R8/R7



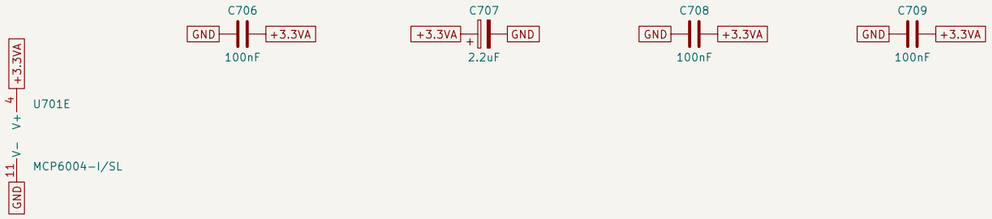
Band-pass into the ADC, U1C: R3/C4 high-pass (4.0 Hz), R4/C5 low-pass (723 Hz), gain R4/R3 = 12.2



Mid-supply reference: R1/R2 divider (1.055 V), C2/C3 filter, U1D buffer



Supply: U1 power pins and bypass



One sEMG channel, placed eight times on the root sheet (CH1 to CH8; references R1xx to R8xx by channel).
Ports: VINP / VINN from the electrode pair, BPFOUT to an ESP32-S3 ADC1 pin. Supply +3.3VA (the LP2985 analog rail), GND.
Simulated: gain 134 V/V at 50 to 100 Hz, 110 V/V at 500 Hz; BPFOUT rests at 1.055 V; CMRR >= 48.6 dB at every corner (test_limits.csv).
U1 is KiCad's generic quad op-amp symbol (LM324 pinout, the same as the fitted part); the fitted part is the Value / MPN field.

Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0065: Muscle BioAmp Patchy v0.2 (CERN-OHL-S-2.0)
ambrflow design loop, engine e735a0e; spec semg8-opamp 73a6d4ada293

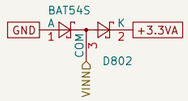
Sheet: /CH7/
File: channel.kicad_sch

Title: sEMG-8 front-end channel (one of eight)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/9

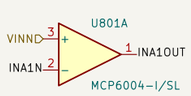
Input: RFI filter and clamps to the rails

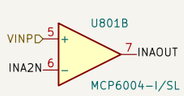






Instrumentation amplifier U1A / U1B: gain 1 + R8/R7 = 11; CMRR needs R5/R6 = R8/R7





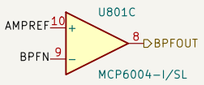








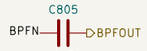
Band-pass into the ADC, U1C: R3/C4 high-pass (4.0 Hz), R4/C5 low-pass (723 Hz), gain R4/R3 = 12.2



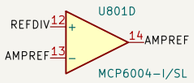




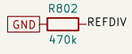


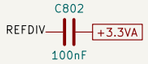


Mid-supply reference: R1/R2 divider (1.055 V), C2/C3 filter, U1D buffer







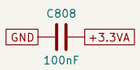




Supply: U1 power pins and bypass











One sEMG channel, placed eight times on the root sheet (CH1 to CH8; references R1xx to R8xx by channel).
Ports: VINP / VINN from the electrode pair, BPFOUT to an ESP32-S3 ADC1 pin. Supply +3.3VA (the LP2985 analog rail), GND.
Simulated: gain 134 V/V at 50 to 100 Hz, 110 V/V at 500 Hz; BPFOUT rests at 1.055 V; CMRR >= 48.6 dB at every corner (test_limits.csv).
U1 is KiCad's generic quad op-amp symbol (LM324 pinout, the same as the fitted part); the fitted part is the Value / MPN field.

Footprints and MPNs: DigiKey BOM 2026-11-02
From vref-0065: Muscle BioAmp Patchy v0.2 (CERN-OHL-S-2.0)
ambrflow design loop, engine e735a0e; spec semg8-opamp 73a6d4ada293

Sheet: /CH8/
File: channel.kicad_sch

Title: sEMG-8 front-end channel (one of eight)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 9/9